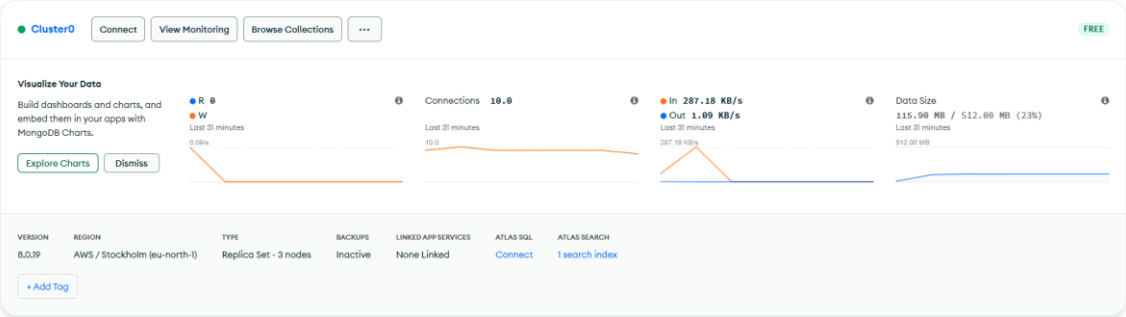
MongoDB

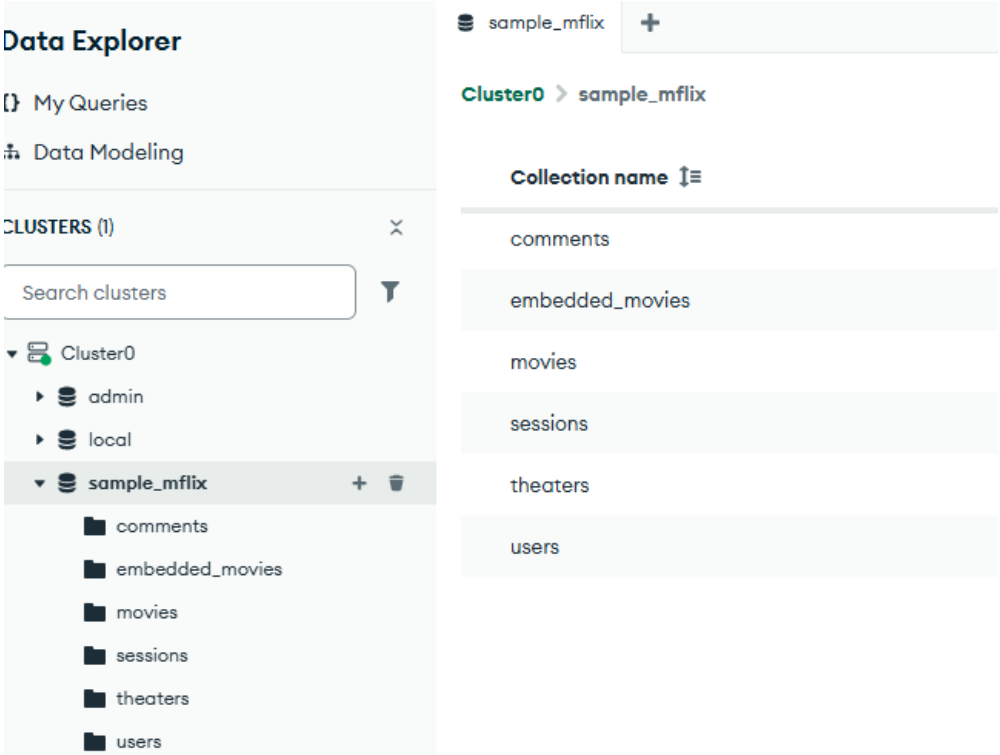
1) APARTADO A

1) Crea un clúster en MongoAtlas, carga los datos de ejemplo sample_training y:

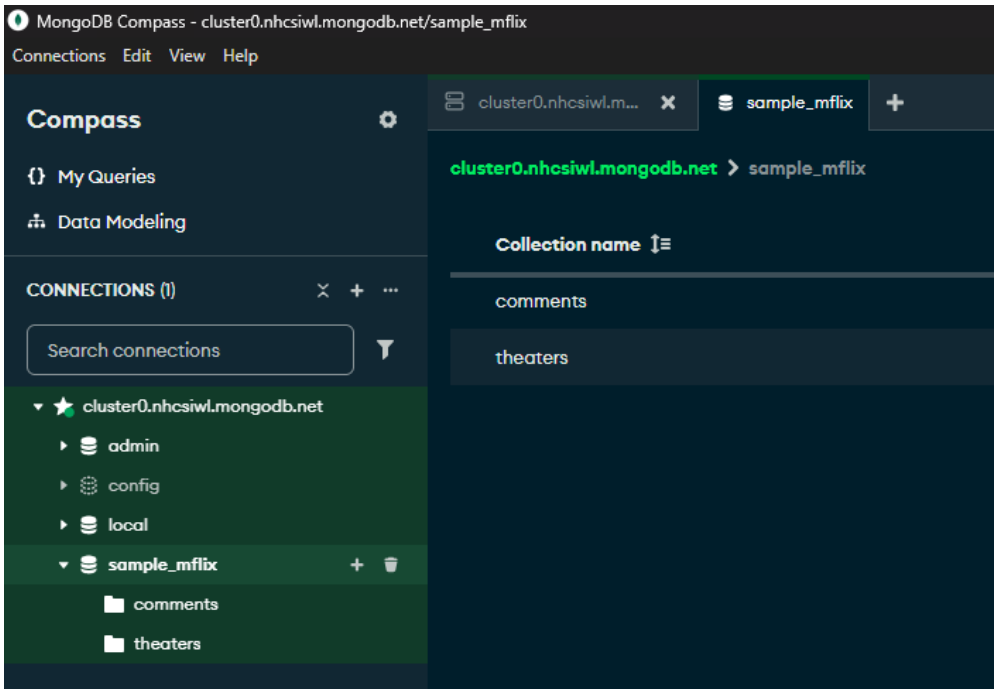
   a) *Dashboard del clúster*



   b) *Bases de datos / colecciones creadas*

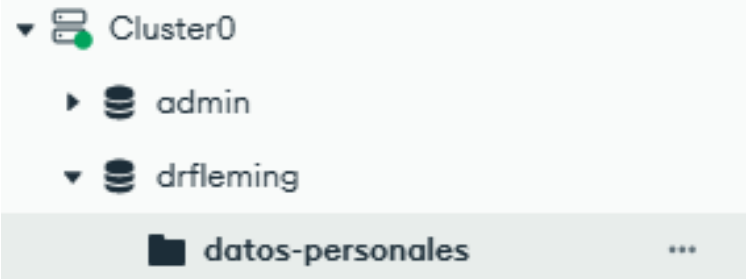


2) Conéctate mediante MongoDB Compass y adjunta una captura de pantalla tras conectar con el clúster.

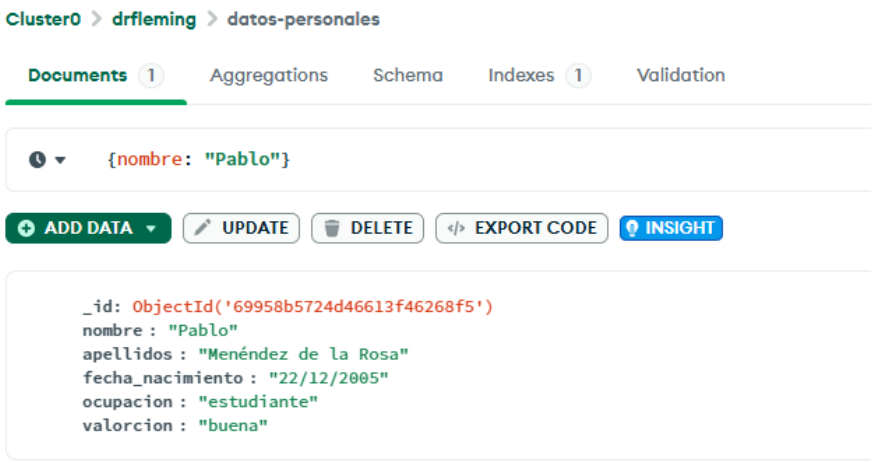


3) En la base de datos drfleming, crea un documento con tus datos personales en la colección alumnos. Tras insertarlo, realiza una consulta para recuperarlo.

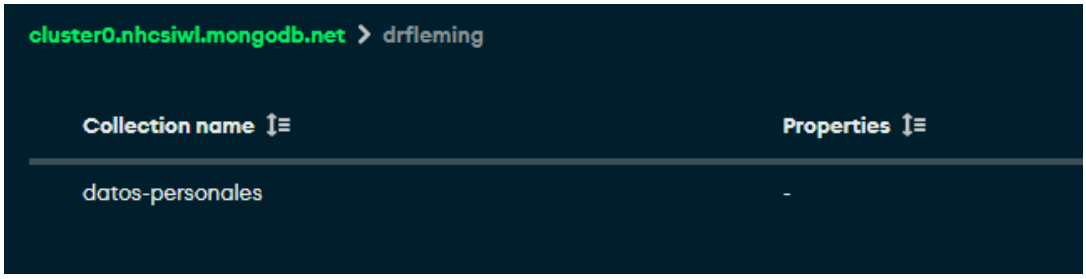
Base de datos:



Datos:



4) Recupera las colecciones existentes en la base de datos.



2) APARTADO B

1) Haciendo uso de mongosh, escribe los comandos necesarios y adjunta el resultado de las consultas para:

a) Obtener las bases de datos creadas.

```
> show dbs
< drfleming            72.00 KiB
  sample_airbnb        60.45 MiB
  sample_analytics     13.60 MiB
  sample_geospatial     1.98 MiB
  sample_guides        72.00 KiB
  sample_mflix        111.67 MiB
  sample_restaurants    9.73 MiB
  sample_supplies       2.02 MiB
  sample_training      65.51 MiB
  sample_weatherdata    4.30 MiB
  admin               360.00 KiB
  local                 2.98 GiB
```

b) Sobre la base de datos sample_training

1. Recuperar todas las colecciones

```
> show collections
< companies
  grades
  inspections
  posts
  routes
  trips
  zips
```

2. Recuperar qué base de datos está activa.

```
> use sample_training
< switched to db sample_training
> db
< sample_training
```

c) *Sobre la colección sample_training.zips:*

1. *Recuperar todos los documentos.*

```
> db.zips.find()
< {
    _id: ObjectId('5c8eccc1caa187d17ca6ed16'),
    city: 'ALPINE',
    zip: '35014',
    loc: {
      y: 33.331165,
      x: 86.208934
    },
    pop: 3062,
    state: 'AL'
  }
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed17'),
    city: 'BESSEMER',
    zip: '35020',
    loc: {
      y: 33.409002,
      x: 86.947547
    },
    pop: 40549,
    state: 'AL'
  }
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed18'),
    city: 'ACMAR',
    zip: '35004',
    loc: {
      y: 33.584132,
      x: 86.51557
    },
    pop: 6055,
    state: 'AL'
  }
```

2. *Recuperar el primer documento.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.findOne()
{
  _id: ObjectId('5c8eccc1caa187d17ca6ed16'),
  city: 'ALPINE',
  zip: '35014',
  loc: { y: 33.331165, x: 86.208934 },
  pop: 3062,
  state: 'AL'
}
```

3. *Recuperar los documentos de la ciudad de SAN DIEGO.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({city: 'SAN DIEGO'})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
    city: 'SAN DIEGO',
    zip: '92101',
    loc: { y: 32.71852, x: 117.159316 },
    pop: 20265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),
    city: 'SAN DIEGO',
    zip: '92102',
    loc: { y: 32.713893, x: 117.121858 },
    pop: 45265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b9'),
    city: 'SAN DIEGO',
    zip: '92103',
    loc: { y: 32.746638, x: 117.163552 },
    pop: 31123,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4bb'),
    city: 'SAN DIEGO',
    zip: '92104',
    loc: { y: 32.745425, x: 117.127189 },
    pop: 44032,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4bc'),
    city: 'SAN DIEGO',
    zip: '92105',
    loc: { y: 32.7423, x: 117.094681 },
    pop: 63344,
    state: 'CA'
  },
```

4. *Recuperar los códigos postales (zip) cuya ciudad sea SAN DIEGO y el estado sea CA.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({city: 'SAN DIEGO', state: 'CA'})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
    city: 'SAN DIEGO',
    zip: '92101',
    loc: { y: 32.71852, x: 117.159316 },
    pop: 20265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),
    city: 'SAN DIEGO',
    zip: '92102',
    loc: { y: 32.713893, x: 117.121858 },
    pop: 45265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b9'),
    city: 'SAN DIEGO',
    zip: '92103',
    loc: { y: 32.746638, x: 117.163552 },
    pop: 31123,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4bb'),
    city: 'SAN DIEGO',
    zip: '92104',
    loc: { y: 32.745425, x: 117.127189 },
    pop: 44032,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4bc'),
    city: 'SAN DIEGO',
    zip: '92105',
    loc: { y: 32.7423, x: 117.094681 },
    pop: 63344,
    state: 'CA'
  },
```

5. *Recuperar los documentos cuya ciudad sea SAN DIEGO pero no pertenezcan al estado de CA.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$nor: [{city: 'SAN DIEGO'}]})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed16'),
    city: 'ALPINE',
    zip: '35014',
    loc: { y: 33.331165, x: 86.208934 },
    pop: 3062,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed17'),
    city: 'BESSEMER',
    zip: '35020',
    loc: { y: 33.409002, x: 86.947547 },
    pop: 40549,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed18'),
    city: 'ACMAR',
    zip: '35004',
    loc: { y: 33.584132, x: 86.51557 },
    pop: 6055,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed19'),
    city: 'BAILEYTON',
    zip: '35019',
    loc: { y: 34.268298, x: 86.621299 },
    pop: 1781,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed1a'),
    city: 'HUEYTOWN',
    zip: '35023',
    loc: { y: 33.414625, x: 86.999607 },
    pop: 39677,
    state: 'AL'
  },
```

6. *Recuperar los documentos cuya ciudad sea SAN DIEGO y el estado sea CA o TX.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$or: [{state: 'CA'}, {state: 'TX'}], city: 'SAN DIEGO'})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
    city: 'SAN DIEGO',
    zip: '92101',
    loc: { y: 32.71852, x: 117.159316 },
    pop: 20265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),
    city: 'SAN DIEGO',
    zip: '92102',
    loc: { y: 32.713893, x: 117.121858 },
    pop: 45265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b9'),
    city: 'SAN DIEGO',
    zip: '92103',
    loc: { y: 32.746638, x: 117.163552 },
    pop: 31123,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4bb'),
    city: 'SAN DIEGO',
    zip: '92104',
    loc: { y: 32.745425, x: 117.127189 },
    pop: 44032,
    state: 'CA'
```

7. *Recuperar los documentos cuya ciudad sea SAN DIEGO o el estado sea TX.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$or: [{city: 'SAN DIEGO'}, {state: 'TX'}]})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
    city: 'SAN DIEGO',
    zip: '92101',
    loc: { y: 32.71852, x: 117.159316 },
    pop: 20265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),
    city: 'SAN DIEGO',
    zip: '92102',
    loc: { y: 32.713893, x: 117.121858 },
    pop: 45265,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4b9'),
    city: 'SAN DIEGO',
    zip: '92103',
    loc: { y: 32.746638, x: 117.163552 },
    pop: 31123,
    state: 'CA'
  },
```

8. *Recuperar los documentos donde el zip sea 92135, 92136 o 92137.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$or: [{zip: '92135'}, {zip: '92136'}, {zip: '92137'}]})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4d7'),
    city: 'SAN DIEGO',
    zip: '92136',
    loc: { y: 32.681585, x: 117.124678 },
    pop: 11750,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f4e9'),
    city: 'SAN DIEGO',
    zip: '92135',
    loc: { y: 32.702482, x: 117.19202 },
    pop: 8122,
    state: 'CA'
  }
]
```

9. *¿Cuál es el código postal de la ciudad de ALLEN que no tiene habitantes (sólo recupera el zip, no nos interesa ningún otro campo, ni el _id)?*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({city: 'ALLEN', pop: 0}, {_id:0, zip: 1})
[ { zip: '36419' } ]
```

2) Escribe los comandos necesarios para realizar las siguientes consultas sobre la colección zips de la base de datos sample_training:

   a) *Recupera los documentos que tienen menos de 50 personas (campo pop).*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({pop: {$lt: 50}})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6eea3'),
    city: 'ALLEN',
    zip: '36419',
    loc: { y: 31.624266, x: 87.66746 },
    pop: 0,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6eebf'),
    city: 'CARLTON',
    zip: '36515',
    loc: { y: 31.322449, x: 87.837793 },
    pop: 30,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef1c'),
    city: 'MORVIN',
    zip: '36762',
    loc: { y: 31.967305, x: 87.972897 },
    pop: 24,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef65'),
    city: 'CHEVAK',
    zip: '99563',
    loc: { y: 61.583982, x: 164.776457 },
    pop: 0,
    state: 'AK'
  },
```

   b) *Recupera los documentos que tienen entre 50 y 100 personas.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({pop: {$lte: 100, $gte: 50}})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef67'),
    city: 'CLARKS POINT',
    zip: '99569',
    loc: { y: 58.84921, x: 158.451241 },
    pop: 68,
    state: 'AK'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef6c'),
    city: 'CHENEGA BAY',
    zip: '99574',
    loc: { y: 60.102558, x: 147.943316 },
    pop: 96,
    state: 'AK'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef71'),
    city: 'EKWOK',
    zip: '99580',
    loc: { y: 59.362792, x: 157.478211 },
    pop: 77,
    state: 'AK'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ef74'),
    city: 'FALSE PASS',
    zip: '99583',
    loc: { y: 54.841028, x: 163.436845 },
    pop: 68,
    state: 'AK'
  },
```

c) *Averigua cuantos documentos hay de la ciudad de SAN DIEGO.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({city: 'SAN DIEGO'}).count()
35
```

d) *Averigua cuantos documentos tienen menos de 100 personas.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({pop: {$lt: 100}}).count()
716
```

e) *Recupera los documentos donde el nombre de la ciudad contenga la palabra BEACH.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({city: {$regex: 'BEACH'}})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6eee2'),
    city: 'ORANGE BEACH',
    zip: '36561',
    loc: { y: 30.290596, x: 87.555888 },
    pop: 2549,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f3ac'),
    city: 'HERMOSA BEACH',
    zip: '90254',
    loc: { y: 33.864309, x: 118.395511 },
    pop: 18289,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f3b3'),
    city: 'MANHATTAN BEACH',
    zip: '90266',
    loc: { y: 33.889594, x: 118.399562 },
    pop: 31984,
    state: 'CA'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f3b5'),
    city: 'REDONDO BEACH',
    zip: '90277',
    loc: { y: 33.830656, x: 118.383221 },
    pop: 33102,
    state: 'CA'
  },
```

f) *Obtén los estados de la ciudad de SAN DIEGO (Solución: ['CA', 'TX']).*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.distinct("state", {city: 'SAN DIEGO'})
[ 'CA', 'TX' ]
```

g) *Listado con los 5 códigos postales más poblados (muestra los documentos completos).*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find().sort({pop: -1}).limit(5)
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca7044d'),
    city: 'CHICAGO',
    zip: '60623',
    loc: { y: 41.849015, x: 87.7157 },
    pop: 112047,
    state: 'IL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca7307f'),
    city: 'BROOKLYN',
    zip: '11226',
    loc: { y: 40.646694, x: 73.956985 },
    pop: 111396,
    state: 'NY'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca72fa0'),
    city: 'NEW YORK',
    zip: '10021',
    loc: { y: 40.768476, x: 73.958805 },
    pop: 106564,
    state: 'NY'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca72fa5'),
    city: 'NEW YORK',
    zip: '10025',
    loc: { y: 40.797466, x: 73.968312 },
    pop: 100027,
    state: 'NY'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6f39d'),
    city: 'BELL GARDENS',
    zip: '90201',
    loc: { y: 33.969177, x: 118.17205 },
    pop: 99568,
    state: 'CA'
  }
]
```

h) *Cantidad de documentos que no tienen menos de 5.000 habitantes ni más de 1.000.000 (debes utilizar el operador $nor).*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$nor: [{pop: {$lt: 5000}, pop: {$gt: 1000000}}]}).count()
29470
```

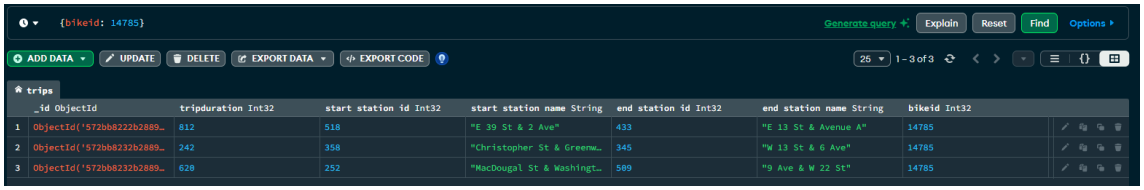
i) *Cuantos documentos tienen más habitantes que su propio código postal (campo zip).*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({$expr:{$gt:['$pop',{$toInt:'$zip'}]}})
[
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed17'),
    city: 'BESSEMER',
    zip: '35020',
    loc: { y: 33.409002, x: 86.947547 },
    pop: 40549,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed1a'),
    city: 'HUEYTOWN',
    zip: '35023',
    loc: { y: 33.414625, x: 86.999607 },
    pop: 39677,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed6a'),
    city: 'BIRMINGHAM',
    zip: '35211',
    loc: { y: 33.481565, x: 86.85904 },
    pop: 35836,
    state: 'AL'
  },
  {
    _id: ObjectId('5c8eccc1caa187d17ca6ed6f'),
    city: 'CENTER POINT',
    zip: '35215',
    loc: { y: 33.635447, x: 86.693197 },
    pop: 43862,
    state: 'AL'
  },
```

3) APARTADO C

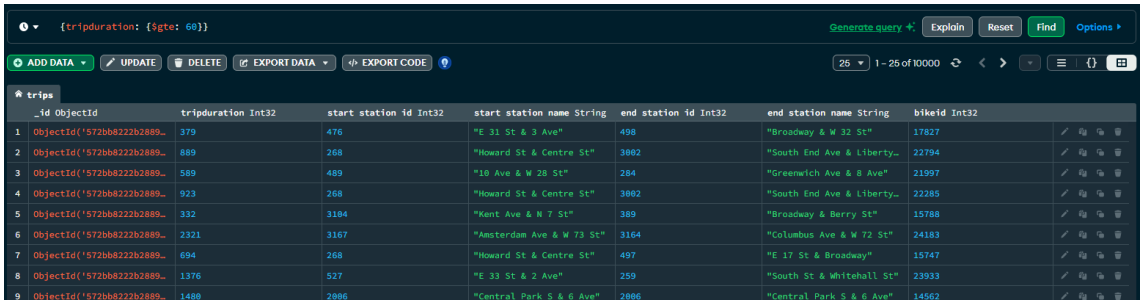
1) Escribe los comandos necesarios para realizar las siguientes consultas sobre la colección trips de la base de datos sample_training:

   a) *Recupera los viajes realizados con la bicicleta con ID 14785*



	_id ObjectId	tripduration Int32	start station id Int32	start station name String	end station id Int32	end station name String	bikeid Int32
1	ObjectId('572bb8222b2889…	812	518	"E 39 St & 2 Ave"	433	"E 13 St & Avenue A"	14785
2	ObjectId('572bb8232b2889…	242	358	"Christopher St & Greenw…	345	"W 13 St & 6 Ave"	14785
3	ObjectId('572bb8232b2889…	620	252	"MacDougal St & Washingt…	509	"9 Ave & W 22 St"	14785

   b) *Encuentra los viajes que duraron más de 1 hora.*



	_id ObjectId	tripduration Int32	start station id Int32	start station name String	end station id Int32	end station name String	bikeid Int32
1	ObjectId('572bb8222b2889…	379	476	"E 31 St & 3 Ave"	498	"Broadway & W 32 St"	17827
2	ObjectId('572bb8222b2889…	889	268	"Howard St & Centre St"	3002	"South End Ave & Liberty…	22794
3	ObjectId('572bb8222b2889…	589	489	"10 Ave & W 28 St"	284	"Greenwich Ave & 8 Ave"	21997
4	ObjectId('572bb8222b2889…	923	268	"Howard St & Centre St"	3002	"South End Ave & Liberty…	22285
5	ObjectId('572bb8222b2889…	332	3104	"Kent Ave & N 7 St"	389	"Broadway & Berry St"	15788
6	ObjectId('572bb8222b2889…	2321	3167	"Amsterdam Ave & W 73 St"	3164	"Columbus Ave & W 72 St"	24183
7	ObjectId('572bb8222b2889…	694	268	"Howard St & Centre St"	497	"E 17 St & Broadway"	15747
8	ObjectId('572bb8222b2889…	1376	527	"E 33 St & 2 Ave"	259	"South St & Whitehall St"	23933
9	ObjectId('572bb8222b2889…	1480	2006	"Central Park S & 6 Ave"	2006	"Central Park S & 6 Ave"	14562

   c) *Mostrar solo el ID de la bicicleta, la duración del viaje y los nombres de las estaciones de los viajes realizados por usuarios nacidos después de 1990.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({'birth year':{ $gt: 1990 }}, {_id: 0, bikeid: 1, tripduration: 1, 'start station name': 1, 'end station name': 1})
[
  {
    tripduration: 694,
    'start station name': 'Howard St & Centre St',
    'end station name': 'E 17 St & Broadway',
    bikeid: 15747
  },
  {
    tripduration: 479,
    'start station name': 'Franklin St & Dupont St',
    'end station name': 'Kent Ave & N 7 St',
    bikeid: 23563
  },
  {
    tripduration: 428,
    'start station name': '6 Ave & W 33 St',
    'end station name': 'W 15 St & 7 Ave',
    bikeid: 23176
  },
```

d) *Recupera los viajes realizados desde la estación Huron St & Franklin St y con duración menor a 500 segundos*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({'start station name': "Huron St & Franklin St", tripduration: {$lt: 500}})
[
  {
    _id: ObjectId('572bb8232b288919b68af1c4'),
    tripduration: 413,
    'start station id': 3116,
    'start station name': 'Huron St & Franklin St',
    'end station id': 3104,
    'end station name': 'Kent Ave & N 7 St',
    bikeid: 16757,
    usertype: 'Subscriber',
    'birth year': 1978,
    'start station location': { type: 'Point', coordinates: [ -73.95826, 40.73266 ] },
    'end station location': { type: 'Point', coordinates: [ -73.96150225, 40.72057658 ] },
    'start time': ISODate('2016-01-02T10:18:16.000Z'),
    'stop time': ISODate('2016-01-02T10:25:09.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b14f6'),
    tripduration: 480,
    'start station id': 3116,
    'start station name': 'Huron St & Franklin St',
    'end station id': 3119,
    'end station name': 'Vernon Blvd & 50 Ave',
    bikeid: 23457,
    usertype: 'Subscriber',
    'birth year': 1978,
    'start station location': { type: 'Point', coordinates: [ -73.95826, 40.73266 ] },
    'end station location': { type: 'Point', coordinates: [ -73.95411749, 40.74232744 ] },
    'start time': ISODate('2016-01-02T17:23:17.000Z'),
    'stop time': ISODate('2016-01-02T17:31:17.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b1566'),
    tripduration: 243,
    'start station id': 3116,
    'start station name': 'Huron St & Franklin St',
    'end station id': 3117,
    'end station name': 'Franklin St & Dupont St',
    bikeid: 16834,
    usertype: 'Subscriber',
    'birth year': 1963,
    'start station location': { type: 'Point', coordinates: [ -73.95826, 40.73266 ] },
    'end station location': { type: 'Point', coordinates: [ -73.95866, 40.73564 ] },
    'start time': ISODate('2016-01-02T17:28:47.000Z'),
    'stop time': ISODate('2016-01-02T17:32:51.000Z')
  }
]
```

e) *Encontrar los 10 viajes más largos, ordenados de mayor a menor duración.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find().sort({tripduration: -1}).limit(10)
[
  {
    _id: ObjectId('572bb8222b288919b68ac07c'),
    tripduration: 326222,
    'start station id': 391,
    'start station name': 'Clark St & Henry St',
    'end station id': 310,
    'end station name': 'State St & Smith St',
    bikeid: 18591,
    usertype: 'Subscriber',
    'birth year': 1979,
    'start station location': { type: 'Point', coordinates: [ -73.99344559, 40.69760127 ] },
    'end station location': { type: 'Point', coordinates: [ -73.98912867, 40.68926942 ] },
    'start time': ISODate('2016-01-01T00:58:20.000Z'),
    'stop time': ISODate('2016-01-04T19:35:23.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b0f0d'),
    tripduration: 279620,
    'start station id': 3165,
    'start station name': 'Central Park West & W 72 St',
    'end station id': 3019,
    'end station name': 'NYCBS Depot - DEL',
    bikeid: 17547,
    usertype: 'Customer',
    'birth year': '',
    'start station location': {
      type: 'Point',
      coordinates: [ -73.9762057363987, 40.77579376683666 ]
    },
    'end station location': { type: 'Point', coordinates: [ -73.981933, 40.716633 ] },
    'start time': ISODate('2016-01-02T16:07:26.000Z'),
    'stop time': ISODate('2016-01-05T21:47:46.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b0593'),
    tripduration: 173357,
    'start station id': 3155,
    'start station name': 'Lexington Ave & E 63 St',
    'end station id': 3083,
    'end station name': 'Bushwick Ave & Powers St',
    bikeid: 15881,
    usertype: 'Customer',
    'birth year': '',
    'start station location': { type: 'Point', coordinates: [ -73.96648977, 40.76440023 ] },
    'end station location': { type: 'Point', coordinates: [ -73.94100005, 40.71247661 ] },
    'start time': ISODate('2016-01-02T14:25:36.000Z'),
    'stop time': ISODate('2016-01-04T14:34:53.000Z')
  },
```

f) Encontrar viajes con estaciones que contengan Ave en su nombre.

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({$or: [{'start station name': {$regex: "Ave"}}, {'end station name': {$regex: "Ave"}}]})
[
  {
    _id: ObjectId('572bb8222b288919b68abf5a'),
    tripduration: 379,
    'start station id': 476,
    'start station name': 'E 31 St & 3 Ave',
    'end station id': 498,
    'end station name': 'Broadway & W 32 St',
    bikeid: 17827,
    usertype: 'Subscriber',
    'birth year': 1969,
    'start station location': { type: 'Point', coordinates: [ -73.97966069, 40.74394314 ] },
    'end station location': { type: 'Point', coordinates: [ -73.98808416, 40.74854862 ] },
    'start time': ISODate('2016-01-01T00:00:45.000Z'),
    'stop time': ISODate('2016-01-01T00:07:04.000Z')
  },
  {
    _id: ObjectId('572bb8222b288919b68abf5b'),
    tripduration: 889,
    'start station id': 268,
    'start station name': 'Howard St & Centre St',
    'end station id': 3002,
    'end station name': 'South End Ave & Liberty St',
    bikeid: 22794,
    usertype: 'Subscriber',
    'birth year': 1961,
    'start station location': { type: 'Point', coordinates: [ -73.99973337, 40.71910537 ] },
    'end station location': { type: 'Point', coordinates: [ -74.015756, 40.711512 ] },
    'start time': ISODate('2016-01-01T00:01:06.000Z'),
    'stop time': ISODate('2016-01-01T00:15:56.000Z')
  },
```

g) *¿Cuántos viajes han realizado los usuarios de tipo Customer?.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({usertype: "Customer"}).count()
1989
```

h) *Recupera los viajes realizados entre el 1 de enero y el 31 de marzo de 2016.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({$or: [{'start time': {$gte:  ISODate('2016-01-01T00:00:00.000Z'), $lte:  ISODate('2016-03-31T23:59:59.000Z')}}, {'start time': {$gte:  ISODa
te('2016-01-01T00:00:00.000Z'), $lte:  ISODate('2016-03-31T23:59:59.000Z')}}]})
[
  {
    _id: ObjectId('572bb8222b288919b68abf5a'),
    tripduration: 379,
    'start station id': 476,
    'start station name': 'E 31 St & 3 Ave',
    'end station id': 498,
    'end station name': 'Broadway & W 32 St',
    bikeid: 17827,
    usertype: 'Subscriber',
    'birth year': 1969,
    'start station location': { type: 'Point', coordinates: [ -73.97966069, 40.74394314 ] },
    'end station location': { type: 'Point', coordinates: [ -73.98808416, 40.74854862 ] },
    'start time': ISODate('2016-01-01T00:00:45.000Z'),
    'stop time': ISODate('2016-01-01T00:07:04.000Z')
  },
  {
    _id: ObjectId('572bb8222b288919b68abf5b'),
    tripduration: 889,
    'start station id': 268,
    'start station name': 'Howard St & Centre St',
    'end station id': 3002,
    'end station name': 'South End Ave & Liberty St',
    bikeid: 22794,
    usertype: 'Subscriber',
    'birth year': 1961,
    'start station location': { type: 'Point', coordinates: [ -73.99973337, 40.71910537 ] },
    'end station location': { type: 'Point', coordinates: [ -74.015756, 40.711512 ] },
    'start time': ISODate('2016-01-01T00:01:06.000Z'),
    'stop time': ISODate('2016-01-01T00:15:56.000Z')
  },
  {
    _id: ObjectId('572bb8222b288919b68abf5c'),
    tripduration: 589,
    'start station id': 489,
    'start station name': '10 Ave & W 28 St',
    'end station id': 284,
    'end station name': 'Greenwich Ave & 8 Ave',
    bikeid: 21997,
    usertype: 'Subscriber',
    'birth year': 1982,
    'start station location': { type: 'Point', coordinates: [ -74.00176882, 40.75066386 ] },
    'end station location': { type: 'Point', coordinates: [ -74.0026376103, 40.7390169121 ] },
    'start time': ISODate('2016-01-01T00:00:48.000Z'),
    'stop time': ISODate('2016-01-01T00:10:37.000Z')
  },
```

i) *Encontrar viajes con duración entre 10 y 20 minutos, hechos por usuarios nacidos antes de 1980.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({ 'tripduration': { $gte: 600, $lte: 1200 }, 'birth year': { lt: 1980 } })
```

j) *Encontrar los viajes que duraron más de una hora o que fueron realizados por usuarios nacidos después del año 2000.*

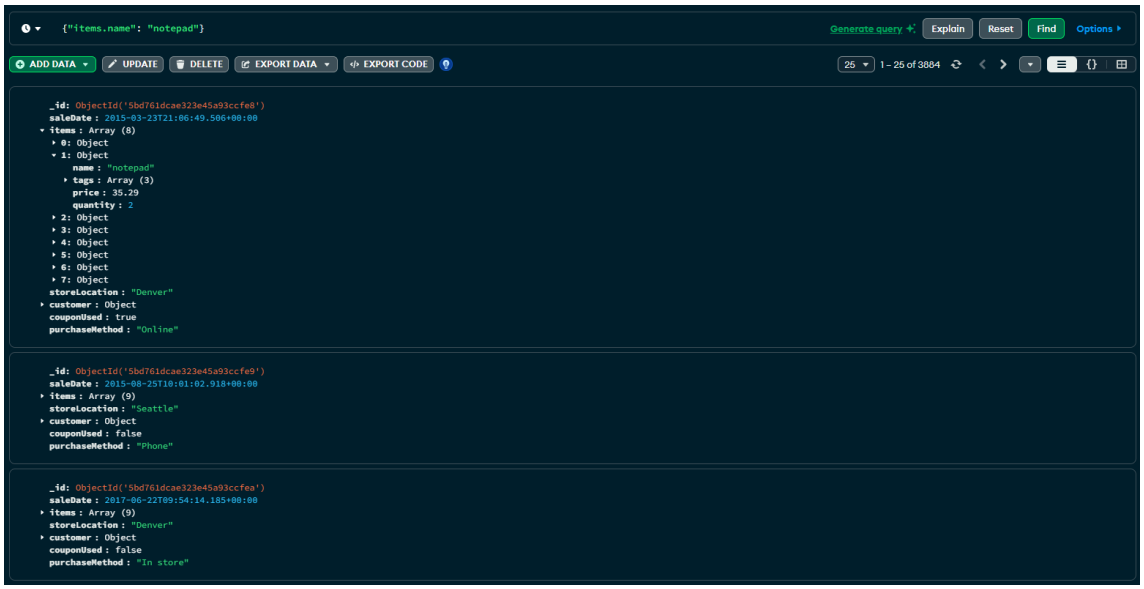
```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find({$or: [{ 'tripduration': { $gt: 600}}, {'birth year': { gt: 2000 } }]})
[
  {
    _id: ObjectId('572bb8222b288919b68abf5b'),
    tripduration: 889,
    'start station id': 268,
    'start station name': 'Howard St & Centre St',
    'end station id': 3002,
    'end station name': 'South End Ave & Liberty St',
    bikeid: 22794,
    usertype: 'Subscriber',
    'birth year': 1961,
    'start station location': { type: 'Point', coordinates: [ -73.99973337, 40.71910537 ] },
    'end station location': { type: 'Point', coordinates: [ -74.015756, 40.711512 ] },
    'start time': ISODate('2016-01-01T00:01:06.000Z'),
    'stop time': ISODate('2016-01-01T00:15:56.000Z')
  },
  {
    _id: ObjectId('572bb8222b288919b68abf5d'),
    tripduration: 923,
    'start station id': 268,
    'start station name': 'Howard St & Centre St',
    'end station id': 3002,
    'end station name': 'South End Ave & Liberty St',
    bikeid: 22285,
    usertype: 'Subscriber',
    'birth year': 1958,
    'start station location': { type: 'Point', coordinates: [ -73.99973337, 40.71910537 ] },
    'end station location': { type: 'Point', coordinates: [ -74.015756, 40.711512 ] },
    'start time': ISODate('2016-01-01T00:00:41.000Z'),
    'stop time': ISODate('2016-01-01T00:16:04.000Z')
  },
  {
    _id: ObjectId('572bb8222b288919b68abf5f'),
    tripduration: 2321,
    'start station id': 3167,
    'start station name': 'Amsterdam Ave & W 73 St',
    'end station id': 3164,
    'end station name': 'Columbus Ave & W 72 St',
    bikeid: 24183,
    usertype: 'Subscriber',
    'birth year': 1963,
    'start station location': {
      type: 'Point',
      coordinates: [ -73.980930447578 42, 40.77966809007312 ]
    },
    'end station location': { type: 'Point', coordinates: [ -73.97898475, 40.7770575 ] },
    'start time': ISODate('2016-01-01T00:01:27.000Z'),
    'stop time': ISODate('2016-01-01T00:40:08.000Z')
  },
```

k) *Mostrar los 5 viajes más recientes, pero saltando los primeros 10 resultados.*

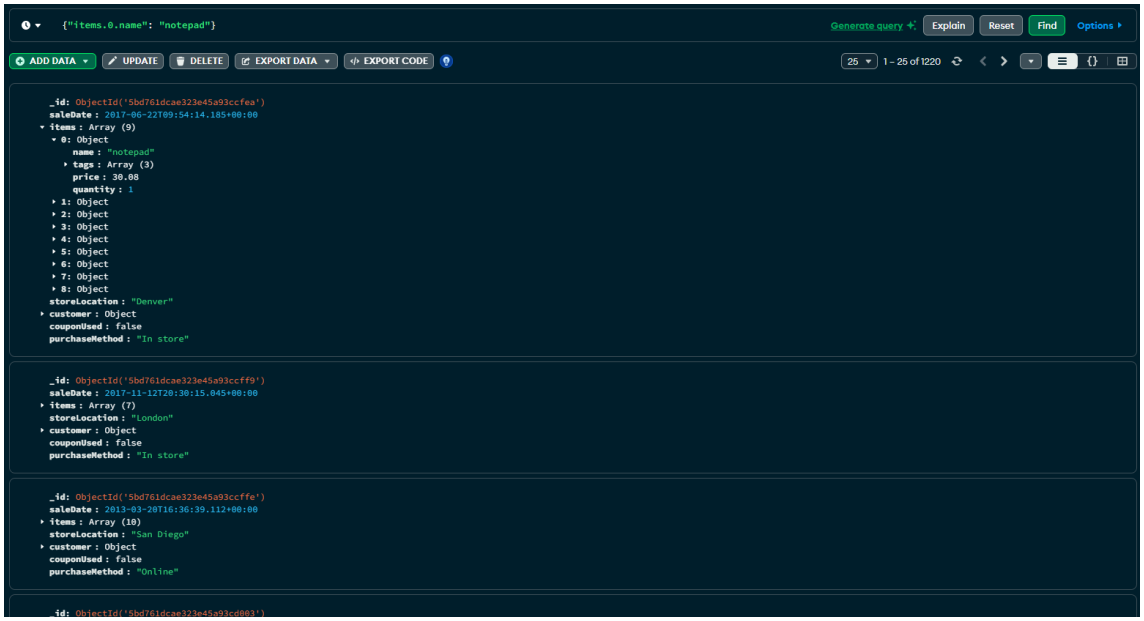
```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.trips.find().sort({'start time': -1}).skip(5).limit(10)
[
  {
    _id: ObjectId('572bb8232b288919b68b1c42'),
    tripduration: 1607,
    'start station id': 212,
    'start station name': 'W 16 St & The High Line',
    'end station id': 494,
    'end station name': 'W 26 St & 8 Ave',
    bikeid: 16554,
    usertype: 'Subscriber',
    'birth year': 1950,
    'start station location': { type: 'Point', coordinates: [ -74.00681753, 40.74334935 ] },
    'end station location': { type: 'Point', coordinates: [ -73.99723551, 40.74734825 ] },
    'start time': ISODate('2016-01-02T19:32:58.000Z'),
    'stop time': ISODate('2016-01-02T19:59:45.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b1c40'),
    tripduration: 3459,
    'start station id': 281,
    'start station name': 'Grand Army Plaza & Central Park S',
    'end station id': 2006,
    'end station name': 'Central Park S & 6 Ave',
    bikeid: 23657,
    usertype: 'Customer',
    'birth year': '',
    'start station location': { type: 'Point', coordinates: [ -73.97371465, 40.7643971 ] },
    'end station location': { type: 'Point', coordinates: [ -73.97634151, 40.76590936 ] },
    'start time': ISODate('2016-01-02T19:32:54.000Z'),
    'stop time': ISODate('2016-01-02T20:30:33.000Z')
  },
  {
    _id: ObjectId('572bb8232b288919b68b1c3f'),
    tripduration: 1262,
    'start station id': 330,
    'start station name': 'Reade St & Broadway',
    'end station id': 504,
    'end station name': '1 Ave & E 15 St',
    bikeid: 23183,
    usertype: 'Subscriber',
    'birth year': 1975,
    'start station location': { type: 'Point', coordinates: [ -74.00562789, 40.71450451 ] },
    'end station location': { type: 'Point', coordinates: [ -73.98165557, 40.73221853 ] },
    'start time': ISODate('2016-01-02T19:32:52.000Z'),
    'stop time': ISODate('2016-01-02T19:53:55.000Z')
  },
```

2) Escribe los comandos necesarios para realizar las siguientes consultas sobre la colección sales de la base de datos sample_supplies:

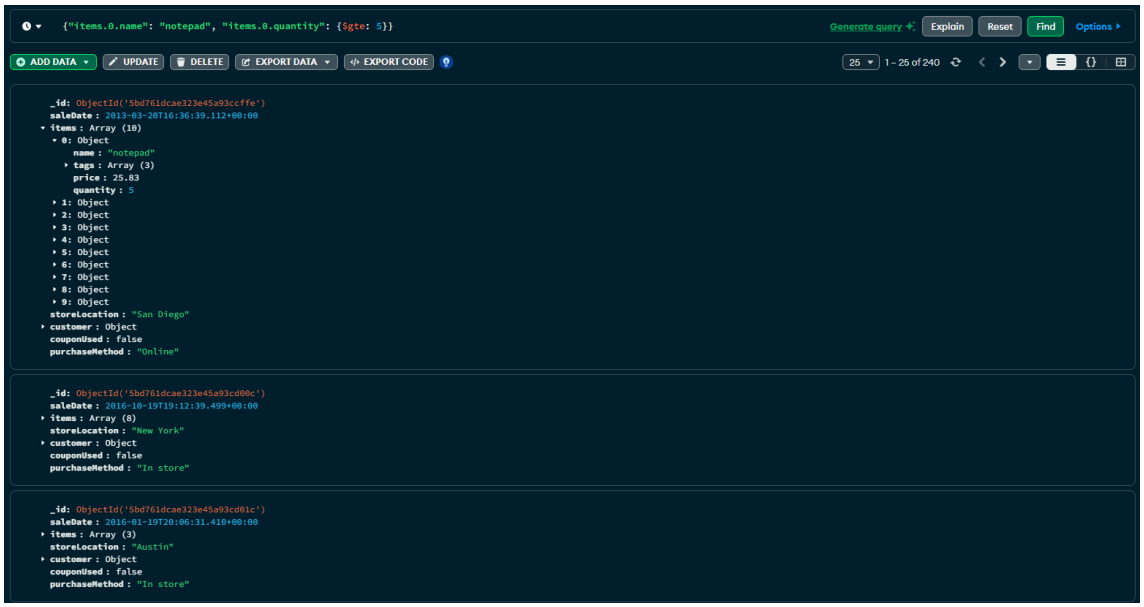
   a) *Recupera los documentos que han vendido algún producto (forma parte del array items) de nombre notepad.*



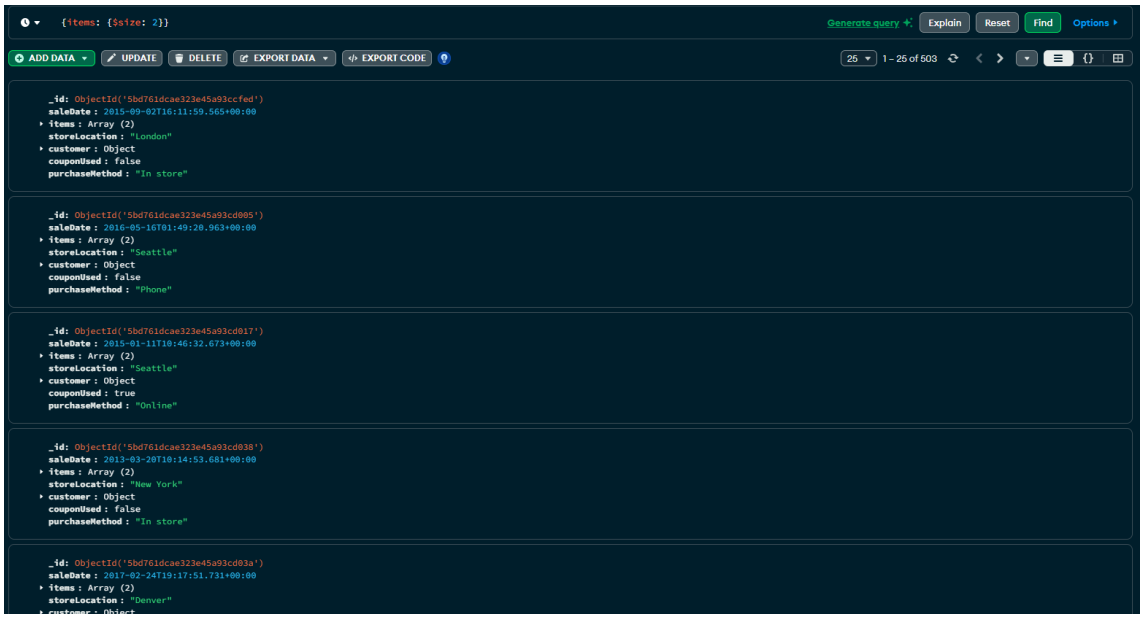
b) *Recupera los documentos donde el primer producto sea notepad.*



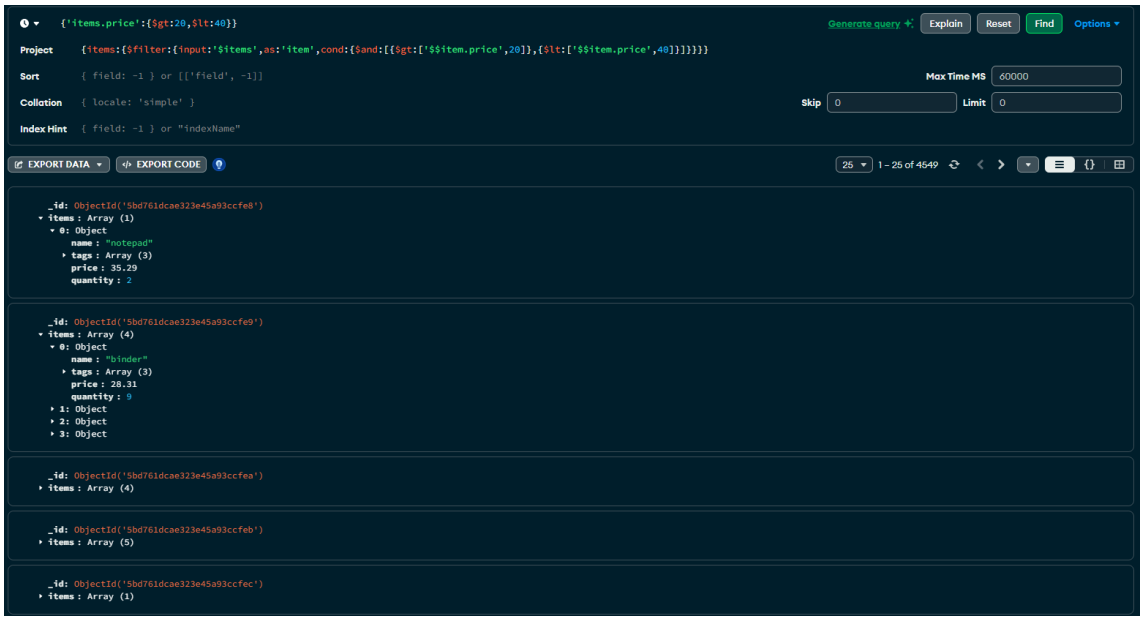
c) *Recupera los documentos donde el primer producto sea notepad y haya vendido un mínimo de 5 (quantity) unidades.*



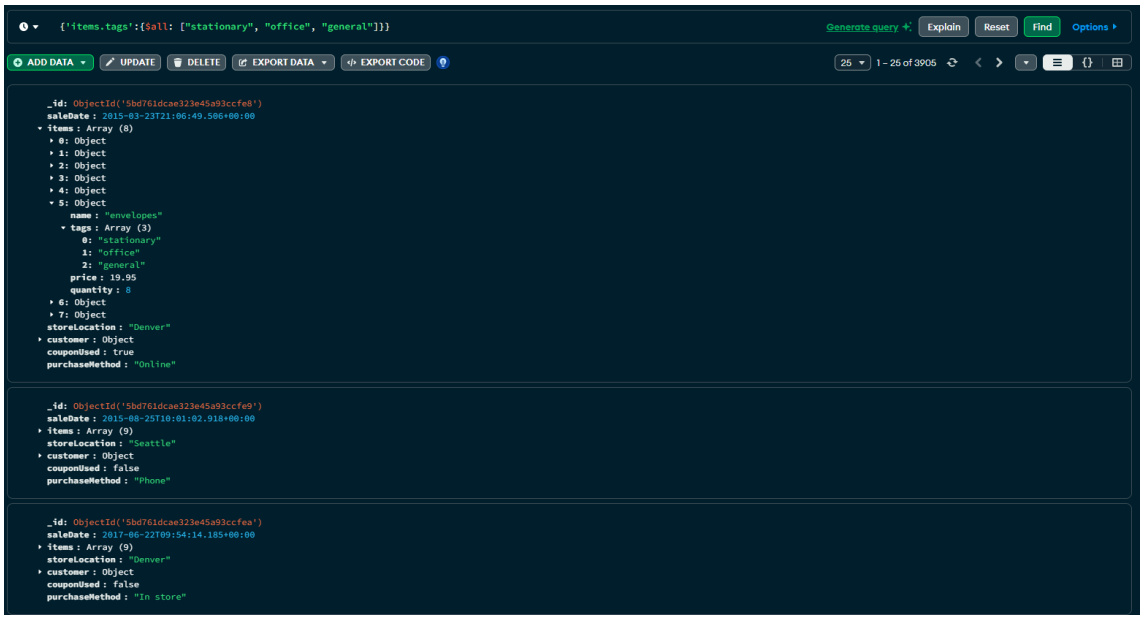
d) *Recupera los documentos que han vendido exactamente dos productos.*



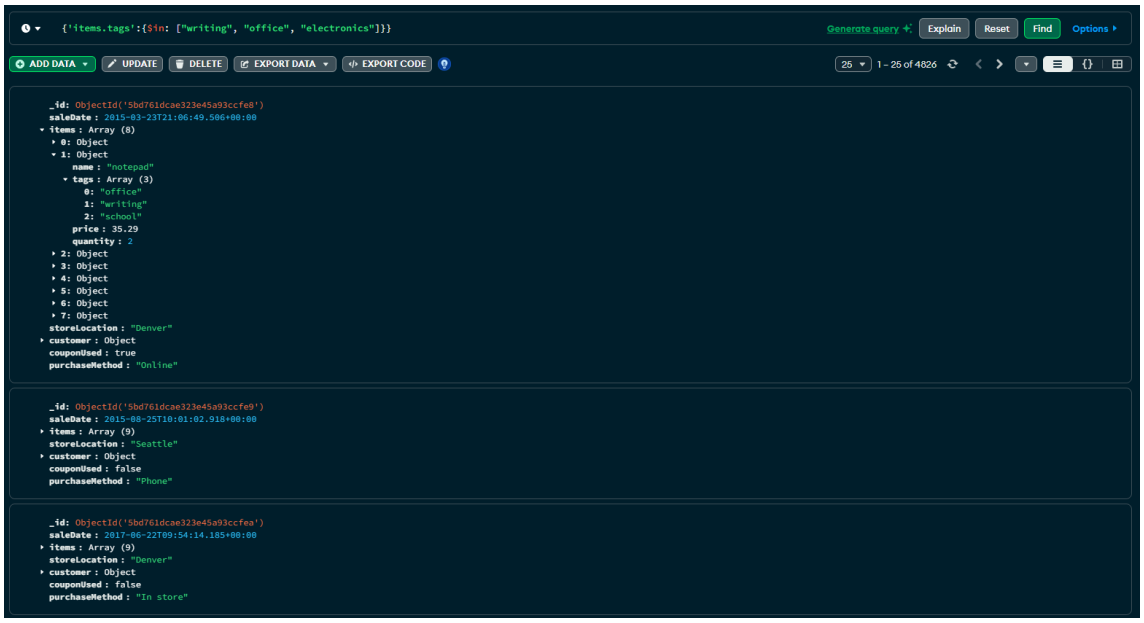
e) *Averiguar las ventas y sus productos donde el precio de los productos es superior a 20€ pero inferior a 40€ (muestra solo los productos que tengan dichos precios).*



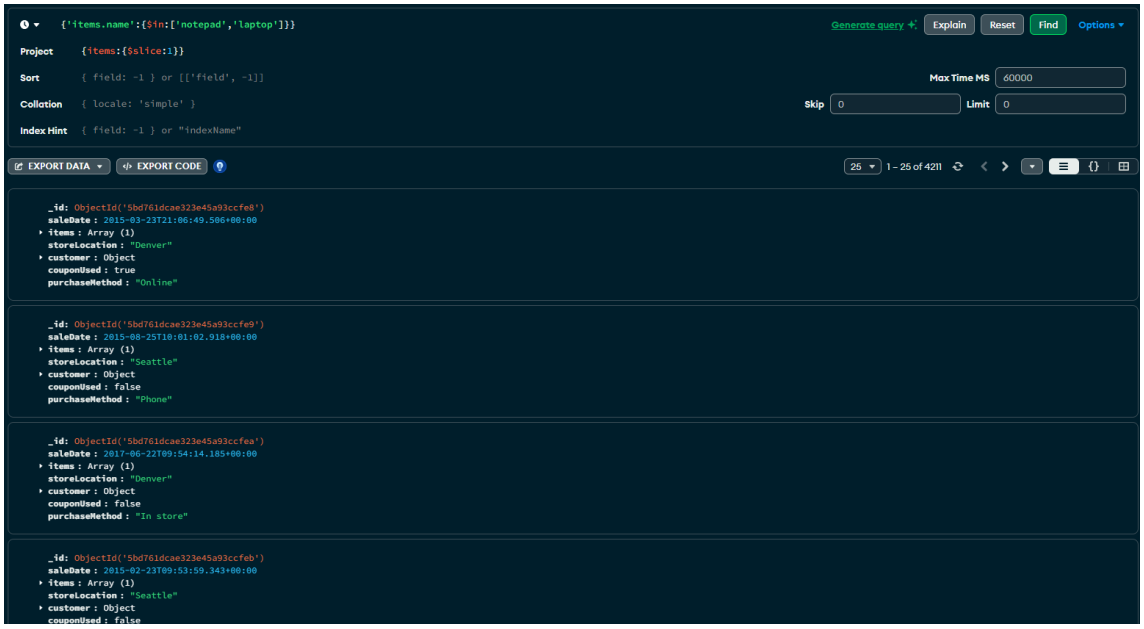
f) *Recupera las ventas donde sus productos tengan las etiquetas stationary, office y general.*



g) *Recupera las ventas donde sus productos tengan las etiquetas writing, office o electronics.*



h) *Para las ventas que sean notepad o laptop, recuperar el primer producto.*



i) *Recupera las ventas realizadas por hombres de más de 40 años que hayan comprado dos productos.*

```
{'items':{$size: 2}, "customer.gender": "M", "customer.age": {$gt: 40}}
```

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

```
_id: ObjectId('5bd761dcae323e45a93ccfed')
saleDate : 2015-09-02T16:11:59.565+00:00
items : Array (2)
storeLocation : "London"
customer : Object
  gender : "M"
  age : 44
  email : "owtar@pu.cd"
  satisfaction : 2
couponUsed : false
purchaseMethod : "In store"

_id: ObjectId('5bd761dcae323e45a93cd005')
saleDate : 2016-05-16T01:49:20.963+00:00
items : Array (2)
storeLocation : "Seattle"
customer : Object
  gender : "M"
  age : 49
  email : "me@lih.st"
  satisfaction : 4
couponUsed : false
purchaseMethod : "Phone"

_id: ObjectId('5bd761dcae323e45a93cd017')
saleDate : 2015-01-11T10:46:32.673+00:00
items : Array (2)
storeLocation : "Seattle"
customer : Object
  gender : "M"
  age : 49
  email : "ce@jinsuod.sy"
  satisfaction : 5
couponUsed : true
purchaseMethod : "Online"

_id: ObjectId('5bd761dcae323e45a93cd055')
saleDate : 2017-04-09T21:55:38.343+00:00
```

j) Para las ventas realizadas por mujeres y realizadas por teléfono (purchaseMethod), recupera dos de los productos adquiridos.

```
{
  "customer.gender": "F",
  purchaseMethod: "Phone"
}
```

Generate query | Explain | Reset | Find | Options

Project { items: { $slice: 2 } }

Sort { field: -1 } or [['field', -1]]

Collation { locale: 'simple' }

Index Hint { field: -1 } or "indexName"

Max Time MS 60000

Skip 0 Limit 0

EXPORT DATA | EXPORT CODE

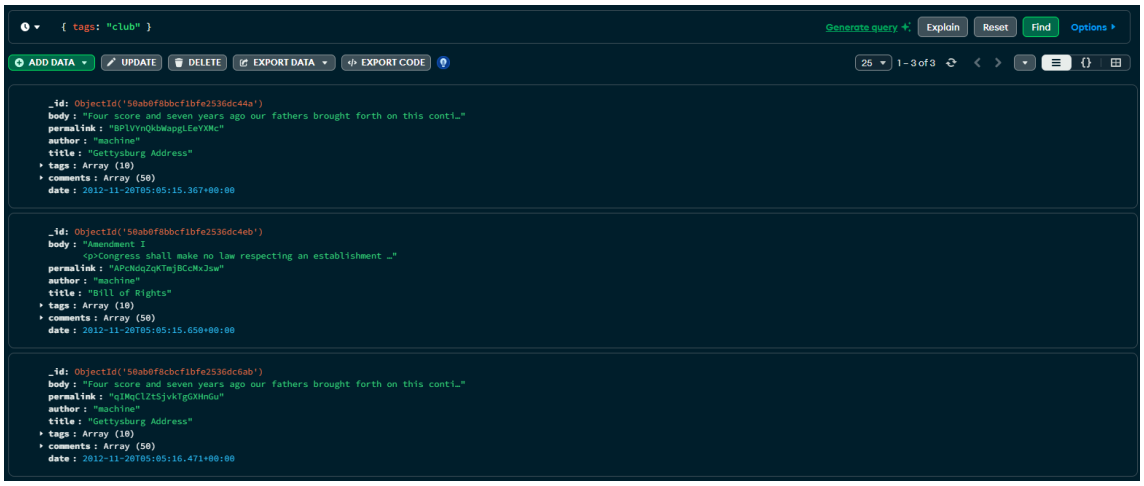
25 | 1 - 25 of 284

```
_id: ObjectId('5bd761dcae323e45a93ccff2')
saleDate : 2015-07-25T07:20:29.804+00:00
items : Array (2)
storeLocation : "Seattle"
customer : Object
  gender : "F"
  age : 34
  email : "keigutip@vaw.tc"
  satisfaction : 4
couponUsed : false
purchaseMethod : "Phone"

_id: ObjectId('5bd761dcae323e45a93ccff8')
saleDate : 2013-08-21T09:36:07.188+00:00
items : Array (2)
storeLocation : "New York"
customer : Object
couponUsed : false
purchaseMethod : "Phone"

_id: ObjectId('5bd761dcae323e45a93cd010')
saleDate : 2016-09-13T16:54:42.141+00:00
items : Array (2)
storeLocation : "San Diego"
customer : Object
couponUsed : false
purchaseMethod : "Phone"
```

4) APARTADO D

1) Escribe los comandos necesarios para realizar las siguientes consultas sobre la colección posts:

a) Recupera los mensajes que tienen la etiqueta club.



b) Recupera los mensajes que tienen la etiqueta club y spy.

c) Recupera los mensajes que tienen la etiqueta club o spy.

```
{ tags: {$in: ["club","spy"] }}
Generate query  Explain  Reset  Find  Options
ADD DATA  UPDATE  DELETE  EXPORT DATA  EXPORT CODE
25  1 - 5 of 5

_id: ObjectId('50ab0f8bbcf1bfe2536dc44a')
body : "Four score and seven years ago our fathers brought forth on this conti…"
permalink : "BPlVYnQkbWapgLEeYXMc"
author : "machine"
title : "Gettysburg Address"
tags : Array (10)
  0: "alibi"
  1: "boundary"
  2: "push"
  3: "club"
  4: "forehead"
  5: "lycra"
  6: "lisa"
  7: "wash"
  8: "hardhat"
  9: "spy"
comments : Array (50)
date : 2012-11-20T05:05:15.367+00:00

_id: ObjectId('50ab0f8bbcf1bfe2536dc4eb')
body : "Amendment I
  <p>Congress shall make no law respecting an establishment …"
permalink : "APcNdqZqKTmjBCcMxJsw"
author : "machine"
title : "Bill of Rights"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.650+00:00

_id: ObjectId('50ab0f8bbcf1bfe2536dc598')
body : "Four score and seven years ago our fathers brought forth on this conti…"
permalink : "wfspPZiEWLrEiwfgbXtM"
author : "machine"
title : "Gettysburg Address"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.959+00:00
```

d) *Cuantos mensajes no tienen la etiqueta restaurant.*

```
{tags:{$ne:'restaurant'}}
Generate query  Explain
ADD DATA  UPDATE  DELETE  EXPORT DATA  EXPORT CODE
25  1 - 25 of 496
```

e) *Cuantos mensajes tienen únicamente un comentario.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.posts.find({comments: {$size: 1}}).count()
0
```

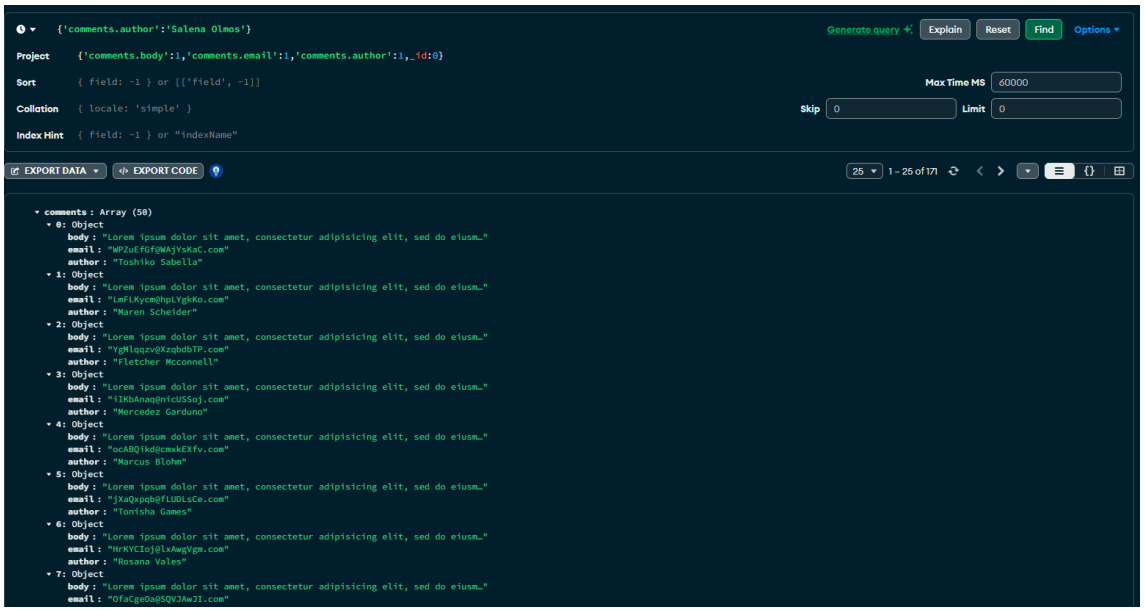
f) *Cuantos mensajes tienen al menos un comentario.*

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.posts.find({'comments.0':{$exists:true}}).count()
500
```

g) *Los mensajes que tienen algún comentario que ha escrito el usuario Salena Olmos.*

```
{ "comments.author": "Salena Olmos" }
Generate query  Explain  Reset  Find  Options
ADD DATA  UPDATE  DELETE  EXPORT DATA  EXPORT CODE
25  1 - 25 of 171

_id: ObjectId('50ab0f8bbcf1bfe2536dc403')
body : "We the People of the United States, in Order to form a more perfect Un…"
permalink : "mzRnhUaDUqVmfZZaPPUy"
author : "machine"
title : "US Constitution"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.246+00:00

_id: ObjectId('50ab0f8bbcf1bfe2536dc406')
body : "We the People of the United States, in Order to form a more perfect Un…"
permalink : "GTSswcmSztqTinKXDADP"
author : "machine"
title : "US Constitution"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.251+00:00

_id: ObjectId('50ab0f8bbcf1bfe2536dc40d')
body : "Four score and seven years ago our fathers brought forth on this conti…"
permalink : "jKFvRIzURrHxfyQUHpPc"
author : "machine"
title : "Gettysburg Address"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.262+00:00

_id: ObjectId('50ab0f8bbcf1bfe2536dc411')
body : "Amendment I
  <p>Congress shall make no law respecting an establishment …"
permalink : "XhawpzKUHQaQhnZIZmeA"
author : "machine"
title : "Bill of Rights"
tags : Array (10)
comments : Array (50)
date : 2012-11-20T05:05:15.269+00:00
```

h) Los comentarios (con su cuerpo y email) que ha escrito el usuario Salena Olmos.



i) Recupera los mensajes que en body contengan la palabra earth, y devuelve el título, 3 comentarios y 5 etiquetas.